

The Nixtamal Guide

The ancestral step that made corn a complete food - and how to bring it back to your kitchen

WHAT NIXTAMALIZATION IS AND WHY IT MATTERS

- Nixtamalization is the process of soaking and cooking dried corn in an alkaline solution (lime water or wood ash water) before grinding.
- The alkali breaks down the pericarp (outer skin), frees bound niacin (vitamin B3), and improves amino acid availability - especially lysine.
- Without it: niacin stays locked in the corn. Populations eating plain corn as a staple developed pellagra (severe niacin deficiency) - skin disease, dementia, death. This happened across Europe and the American South when they adopted corn without adopting the process.
- With it: corn becomes a nutritionally complete food. The Mesoamerican civilizations who invented it 3,500 years ago understood this empirically.

CORN VARIETIES FOR NIXTAMAL

- Best: dried dent corn, dried flint corn, or heirloom varieties (Bloody Butcher, Oaxacan Green, Glass Gem for flour use).
- Avoid: sweet corn (wrong starch structure, turns to mush). Popcorn works but produces a denser masa.
- Source: any heirloom seed supplier, Latin grocery stores (look for 'maiz' or 'masa corn'), or save your own from an open-pollinated variety.

METHOD 1 - CAL (CALCIUM HYDROXIDE / SLAKED LIME)

- Ratio: 1 cup dried corn + 1 tablespoon pickling lime (calcium hydroxide, NOT quicklime) + 3 cups water.
- Combine in a non-reactive pot (stainless, ceramic, or enamel - not aluminium). Bring to a gentle simmer for 20 minutes.
- Remove from heat, cover, let soak 8-12 hours (or overnight). The longer the soak, the softer the masa.
- Rinse thoroughly 3-5 times until water runs clear. The rinsed corn is your nixtamal. Over-processing sign: corn feels slimy; reduce soak time next batch.

METHOD 2 - WOOD ASH WATER

- Soak hardwood ash in water overnight (1 cup ash per 2 litres water), let settle, then pour off the clear grey liquid above the sediment. This is your ash lye.
- Use exactly as you would the cal solution above: same ratio, same soak time.
- Ash water works well and produces a distinctive slightly smoky flavour. Traditional across West Africa, Mesoamerica, and parts of Asia.
- Rinse identically - 3-5 washes until water runs clear.

GRINDING THE MASA

- Traditional: stone metate (volcanic rock). Wet-grind the nixtamal in small batches, adding water as needed until smooth.
- Modern homestead: a cast iron grain mill (Country Living, Diamant, etc.) or a powerful wet grinder (Indian tabletop model works well for masa).
- Texture: masa for tortillas should be smooth, pliable, and not sticky. Add water a tablespoon at a time until you can press

a ball flat without cracking.

MAKING TORTILLAS, MASA HARINA & USING NEJAYOTE

- Tortillas: press golf-ball-sized masa between plastic-lined tortilla press or two cutting boards. Cook 30 seconds each side on a dry cast iron pan at medium-high heat.
- Masa harina (shelf-stable flour): spread masa thin on trays, dry at 60°C for 4-6 hours or air dry 2 days. Crumble, then grind again. Stores for months.
- Nejayote (the soaking liquid): rich in minerals. Dilute 10:1 and use as garden fertiliser, especially for nitrogen-hungry plants.

>> THE SAGE'S TIP: Your first batch will surprise you. The texture of fresh masa is nothing like anything from a bag. Make tortillas the same day and eat them warm.

The step they removed turned a staple into a deficiency. Bring it back to your kitchen. - The Sage

